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D20A-32

Experiment No.: 5
Study of Solidity Program Structure, Basic Constructs, Compilation and Deployment Environment

1. Aim

To study the basic structure and constructs of the Solidity programming language and to create, compile, deploy, and execute smart contracts using the Remix IDE and Remix VM.

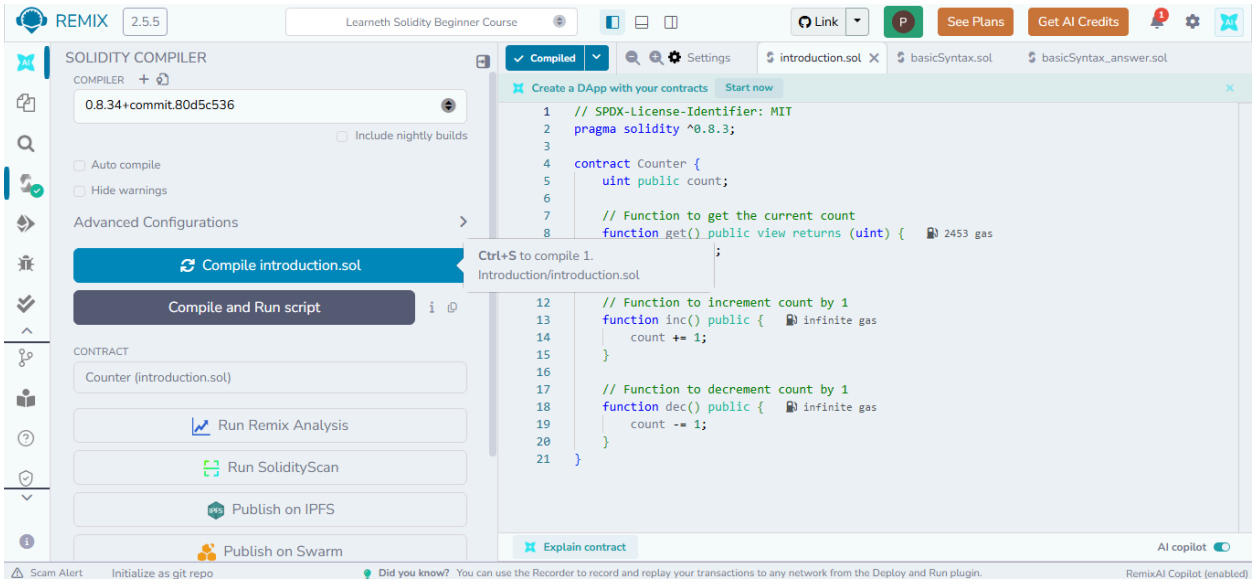
2. Tasks Performed

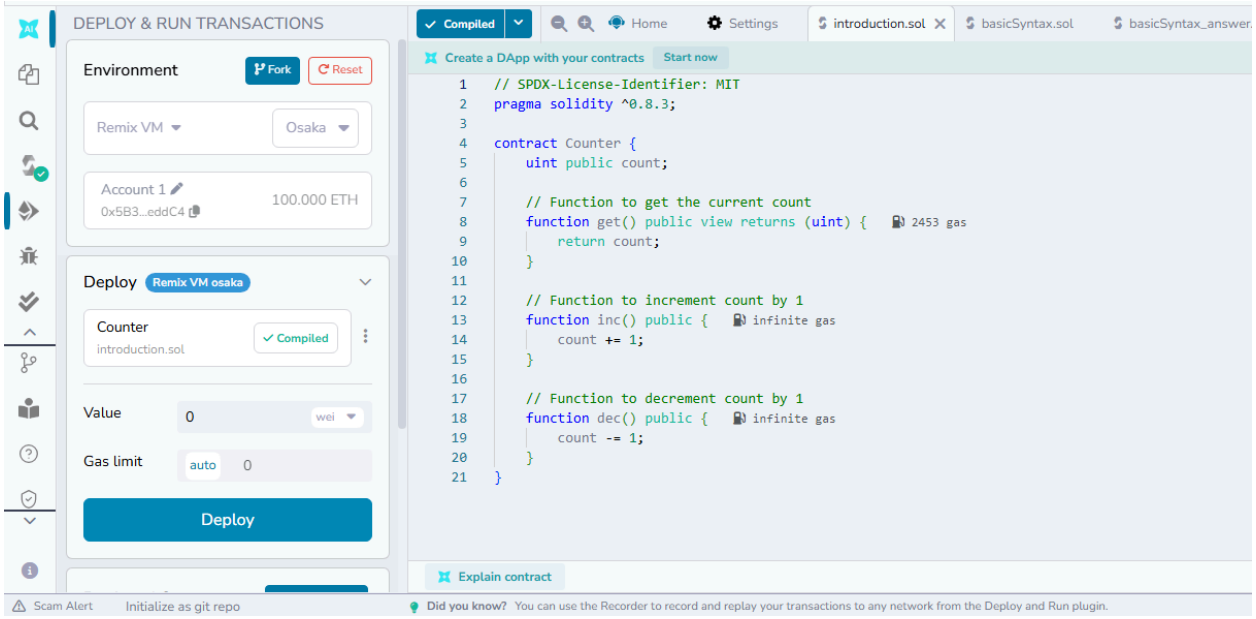
The following tasks were performed during the experiment:

1. Opened and explored the Remix IDE.
2. Explored the Solidity Basics LearnETH course available in Remix IDE.
3. Studied the structure of a Solidity smart contract.
4. Created Solidity programs using:
    - Primitive data types and variables
    - Functions
    - view and pure functions
    - Function inputs and outputs
    - Visibility and modifiers
    - Constructors
    - if-else statements and loops
    - Arrays
    - Mappings
    - Structures and enums
    - Data locations
5. Studied blockchain transaction concepts such as:
    - Ether and Wei
    - Gas and gas price
    - Sending transactions
6. Compiled Solidity programs using the Solidity Compiler in Remix IDE.
7. Deployed smart contracts using the Remix VM.
8. Executed the deployed contract functions and verified their outputs.
9. Completed the 19 assignments provided in the Solidity Basics LearnETH course.

Program Code and Output:

Tutorial -01 Introduction

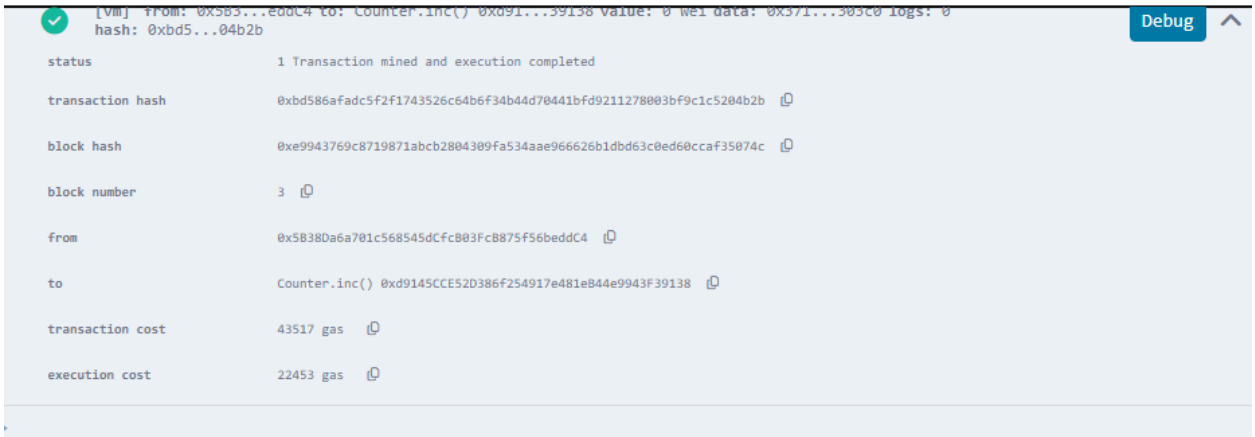




get :



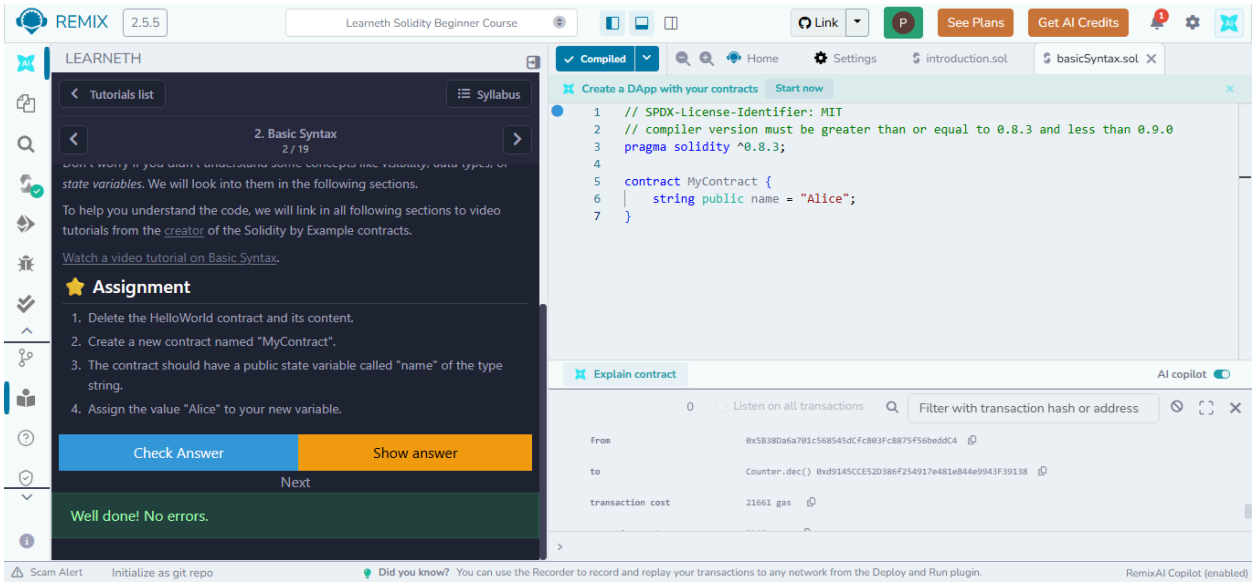
inc:



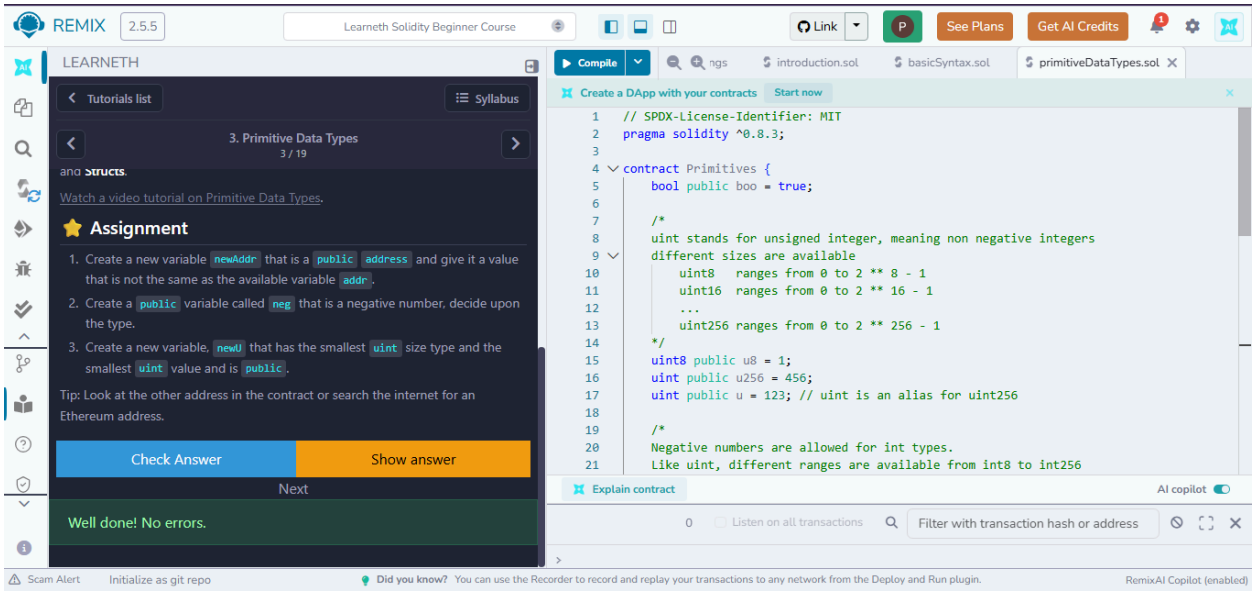
dec:



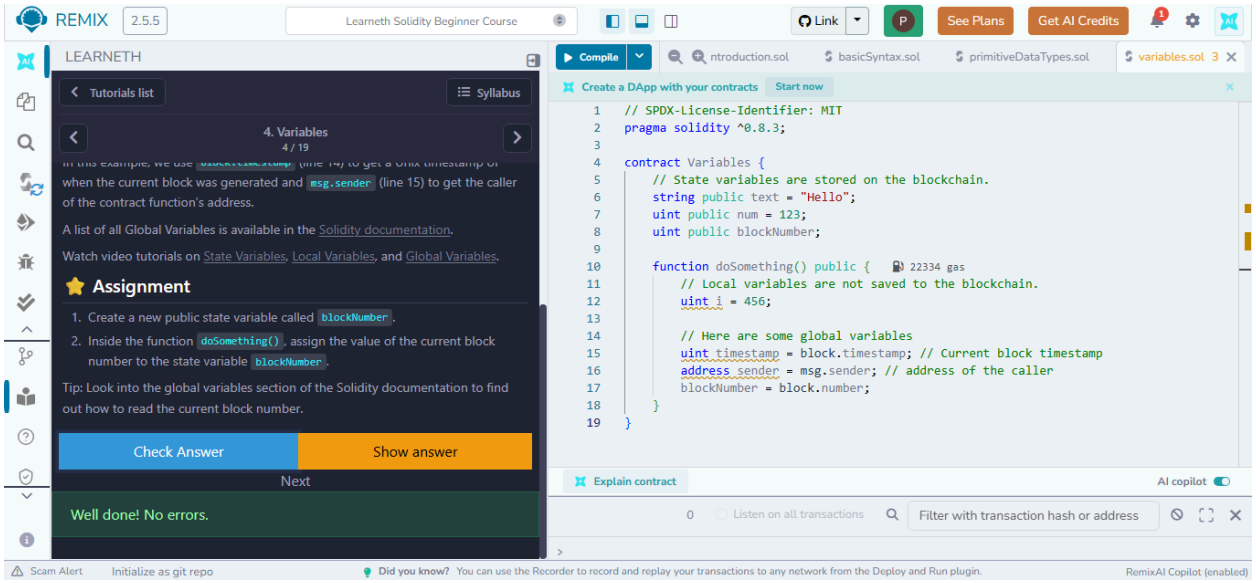
Tutorial-02 Basic Syntax:



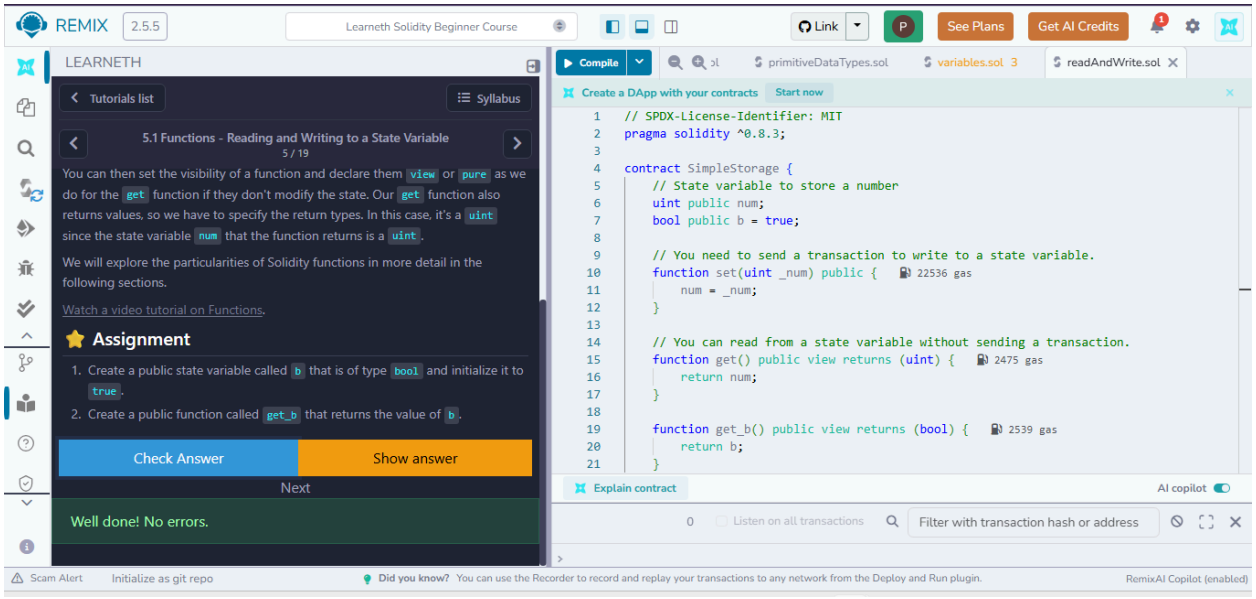
Tutorial-03 Primitive Data Types:



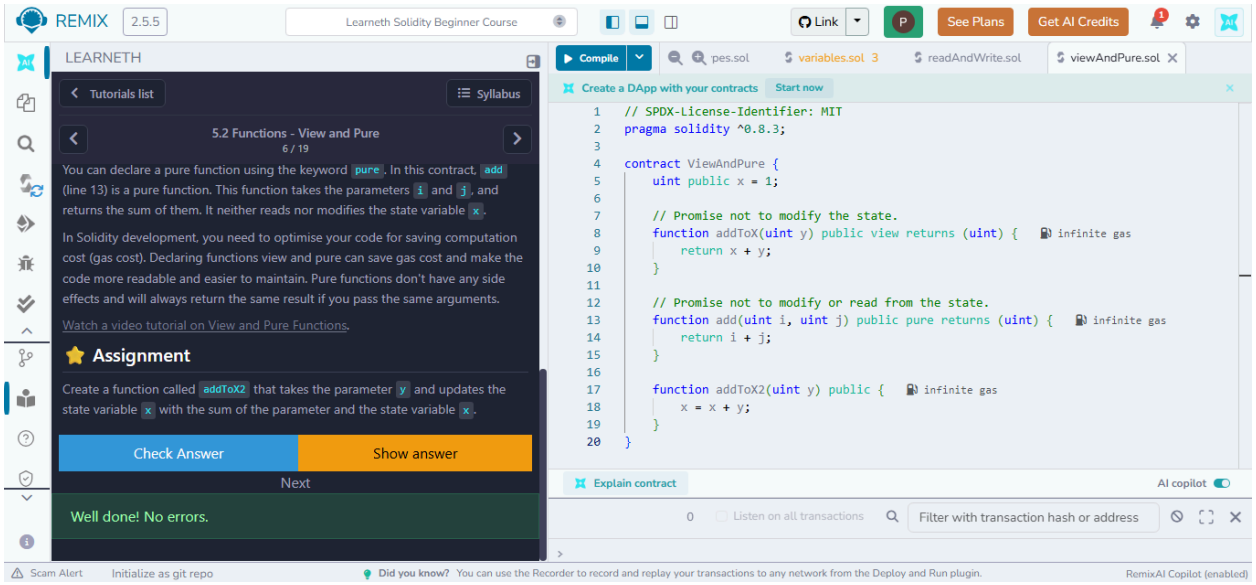
Tutorial-04 Variables:



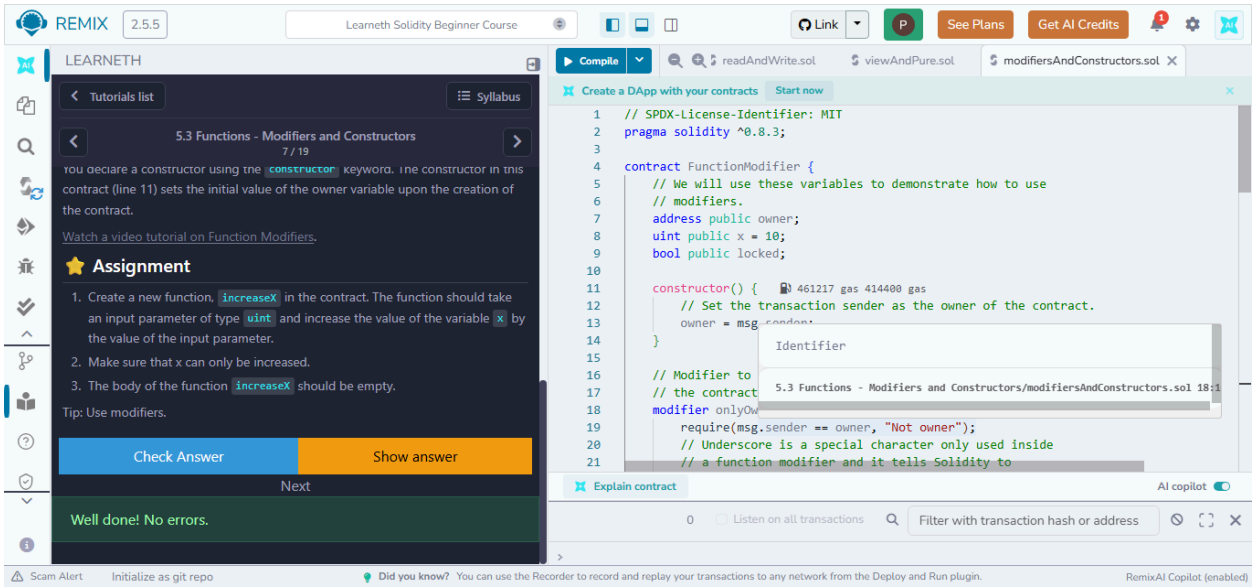
Tutorial-5.1 Reading and Writing to a State Variable :



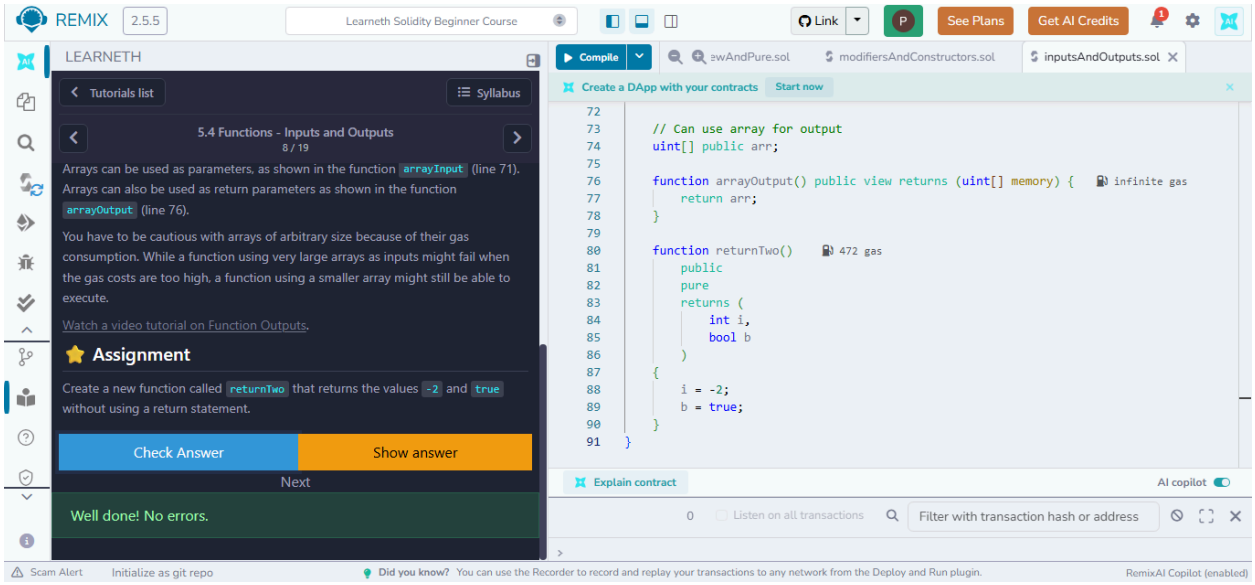
Tutorial-5.2 View and Pure:



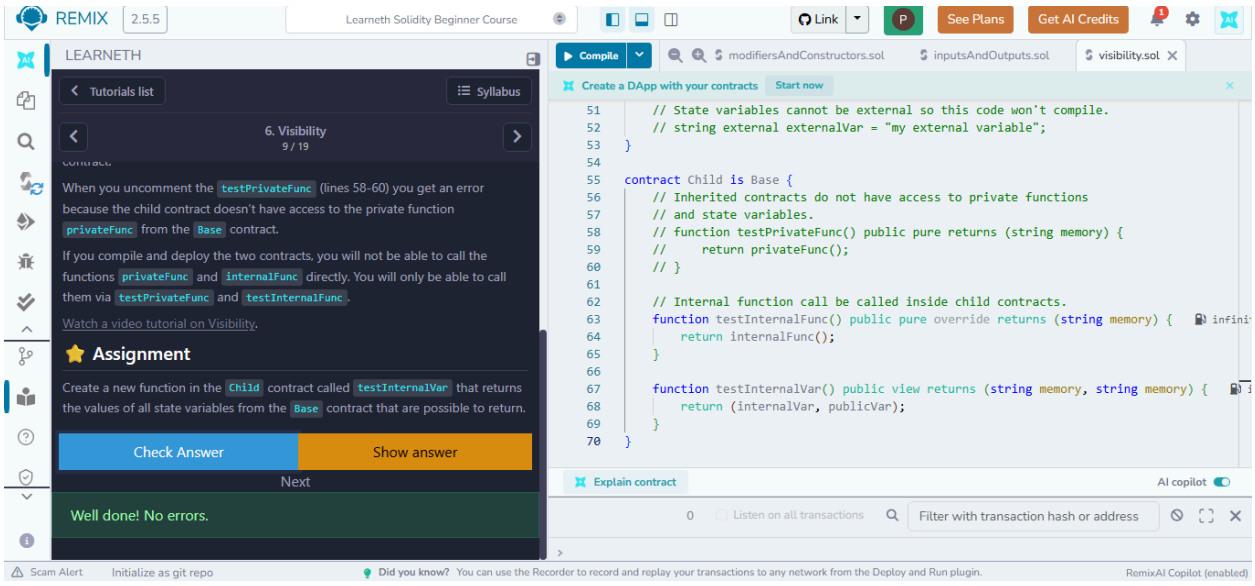
Tutorial-5.3 Functions - Modifiers and Constructors:



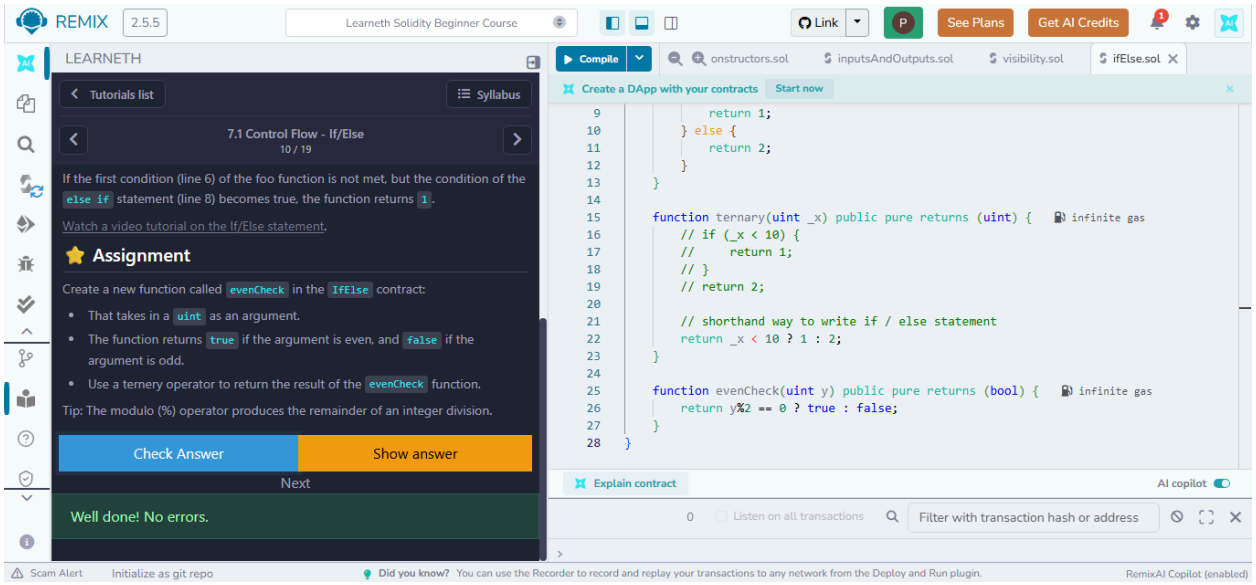
Tutorial-5.4 Functions - Inputs and Outputs:



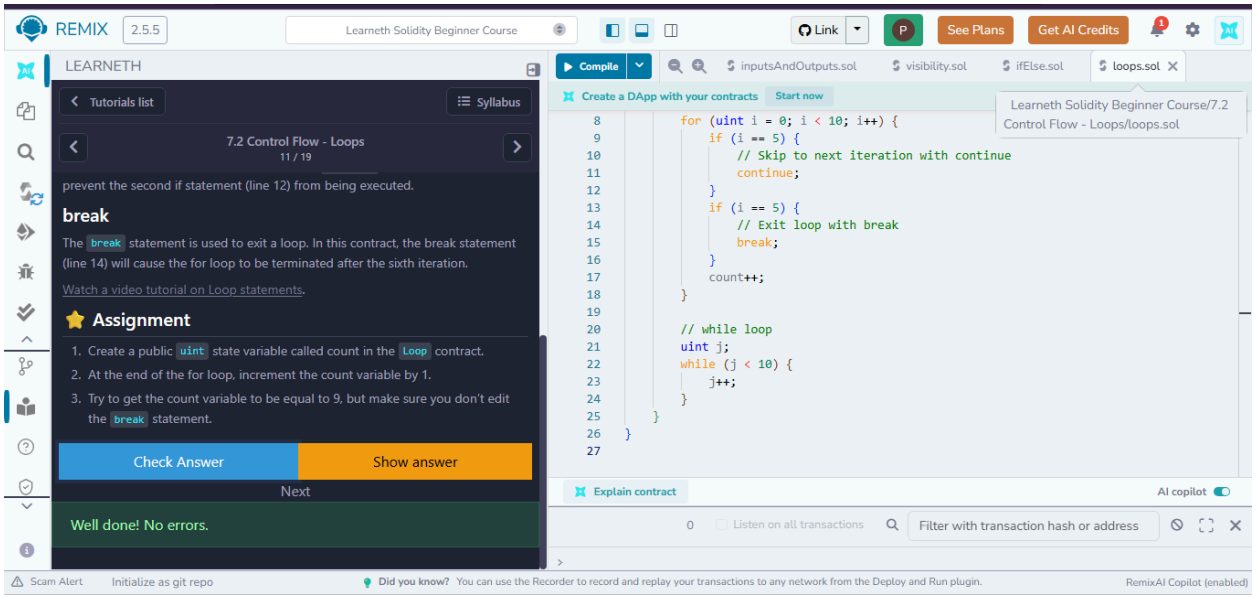
Tutorial-06 Visibility:



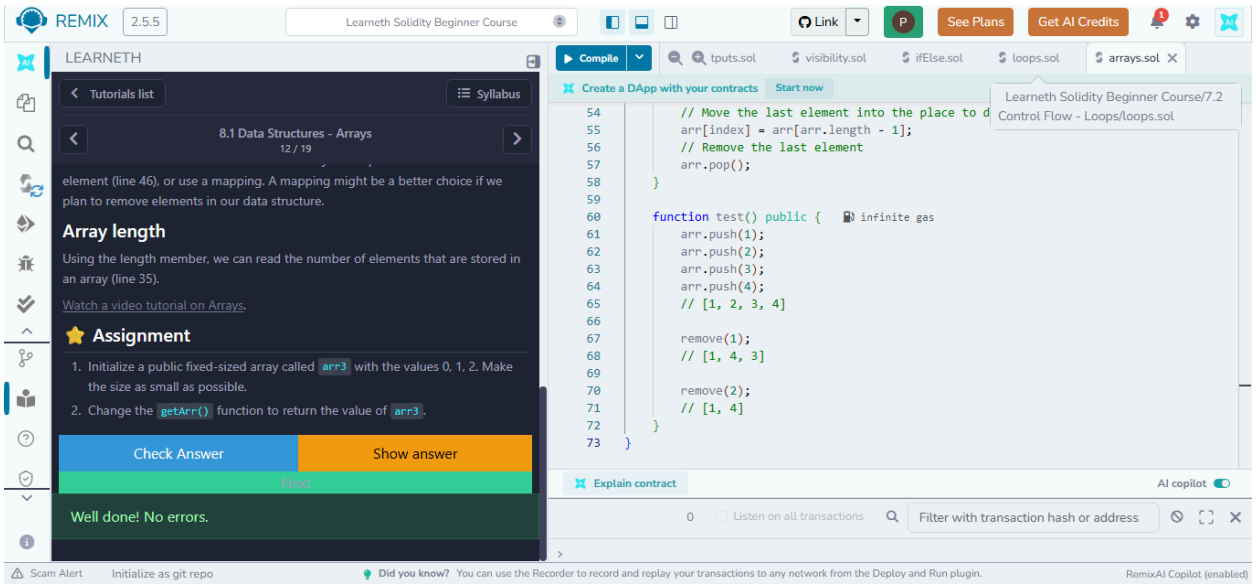
Tutorial-7.1 Control Flow- If/Else:



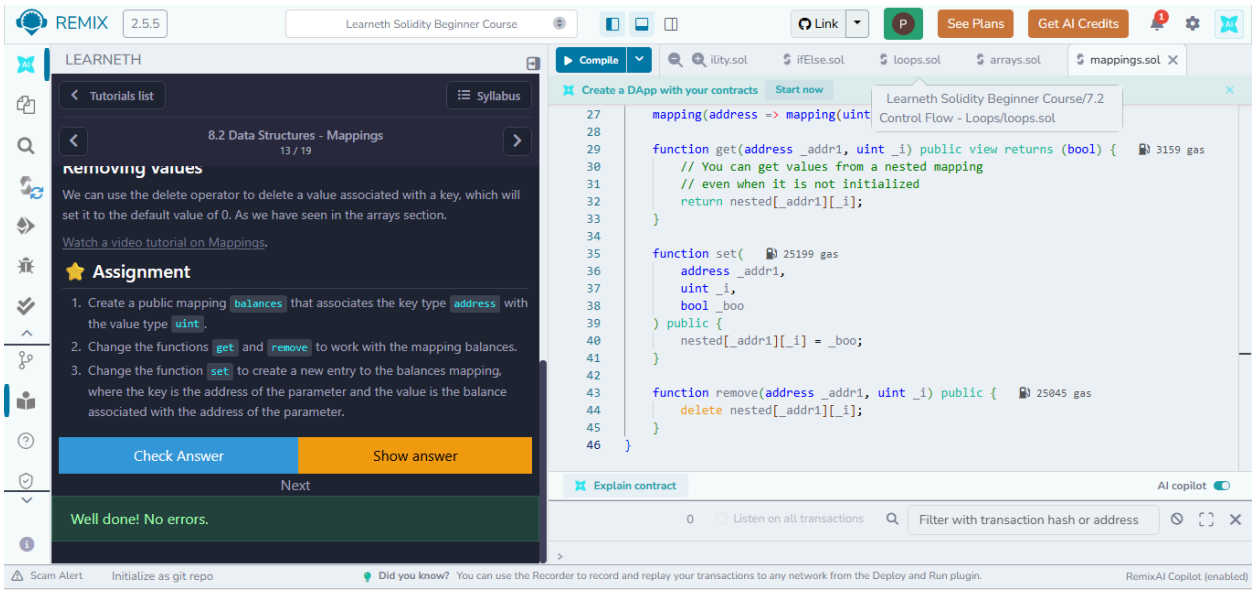
Tutorial-7.2 Control Flow- Loops:



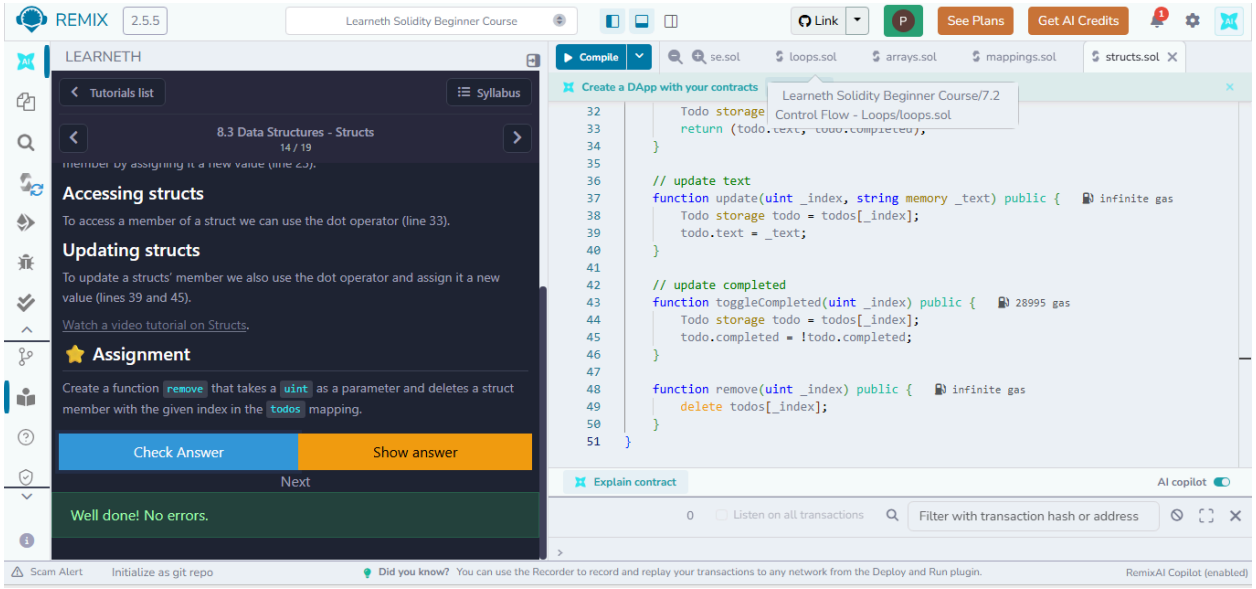
Tutorial-8.1 Data Structures- Arrays :



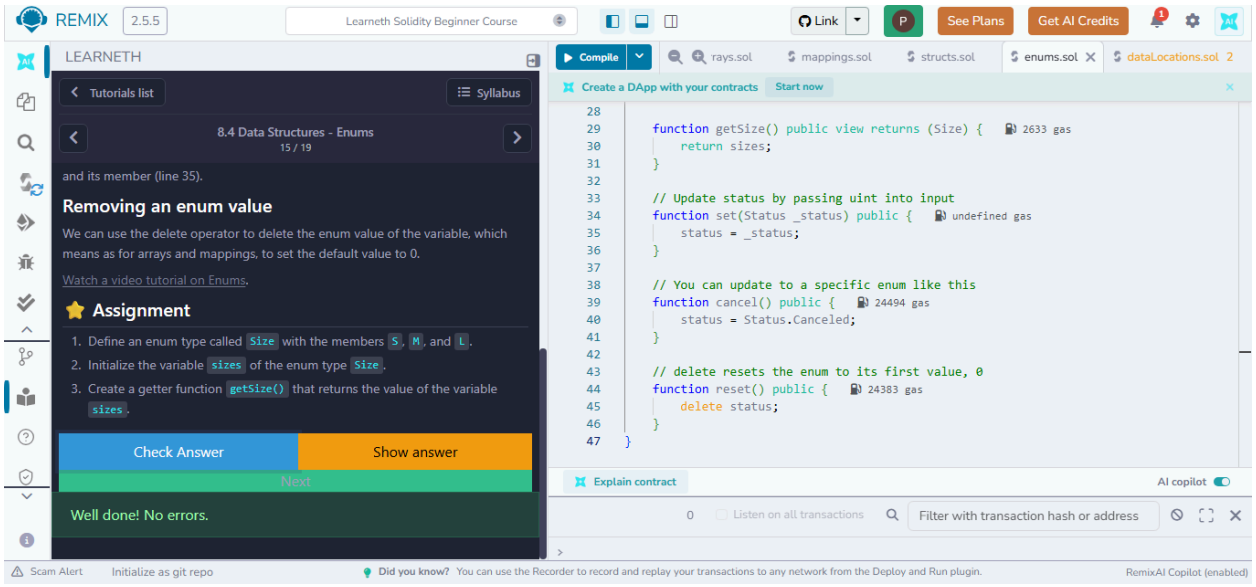
Tutorial-8.2 Data Structures- Mappings:



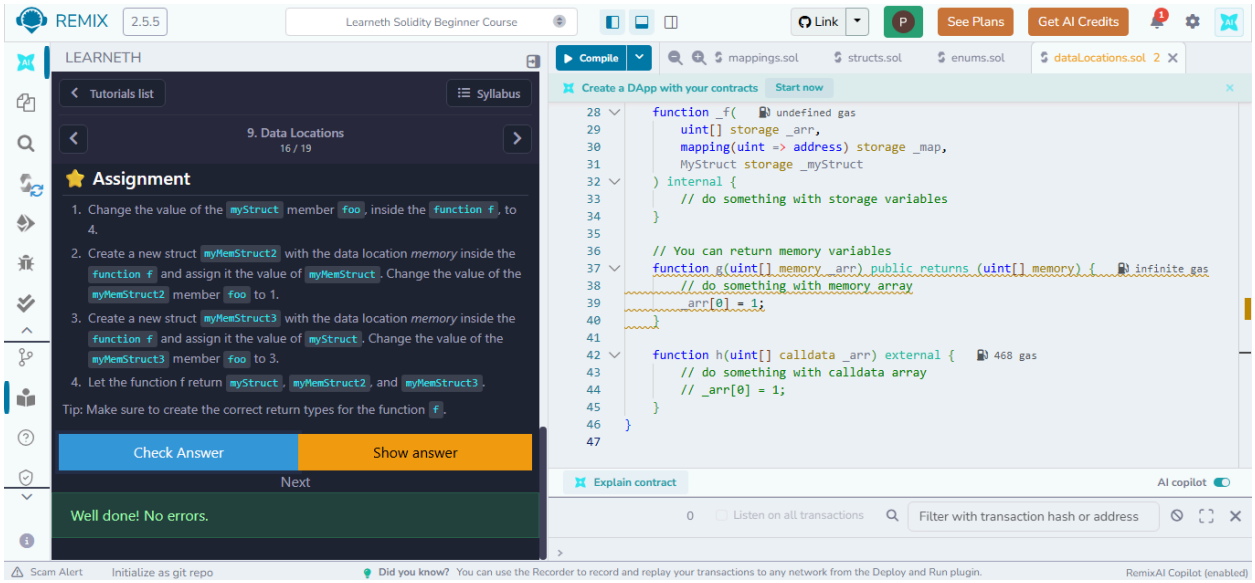
Tutorial-8.3 Data Structures- Structs:



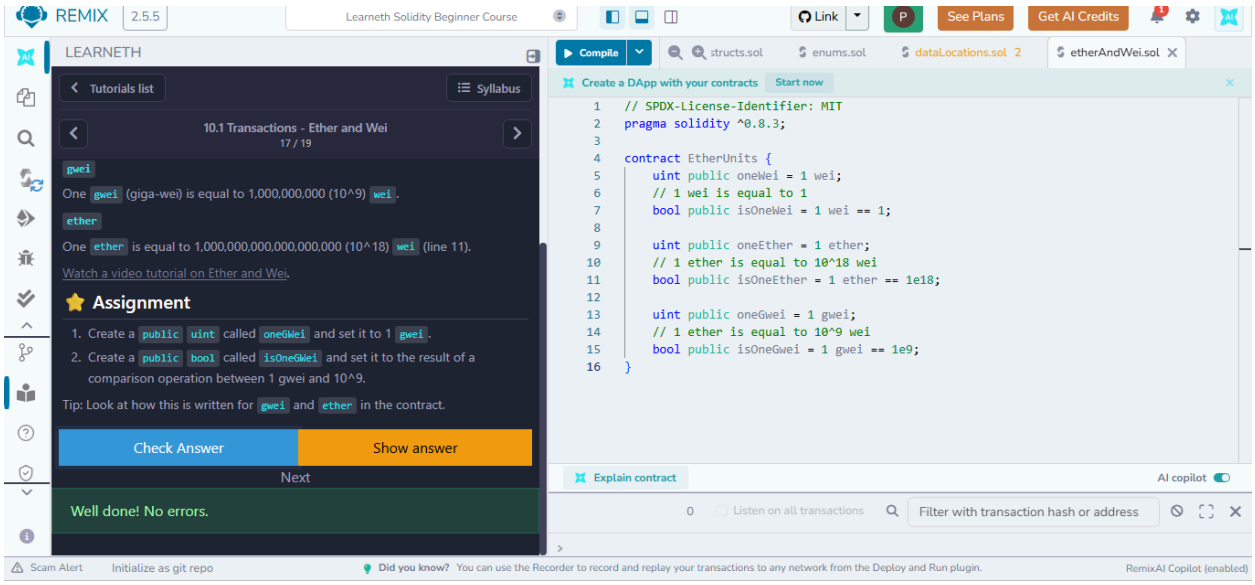
Tutorial-8.4 Data Structures- Enums:



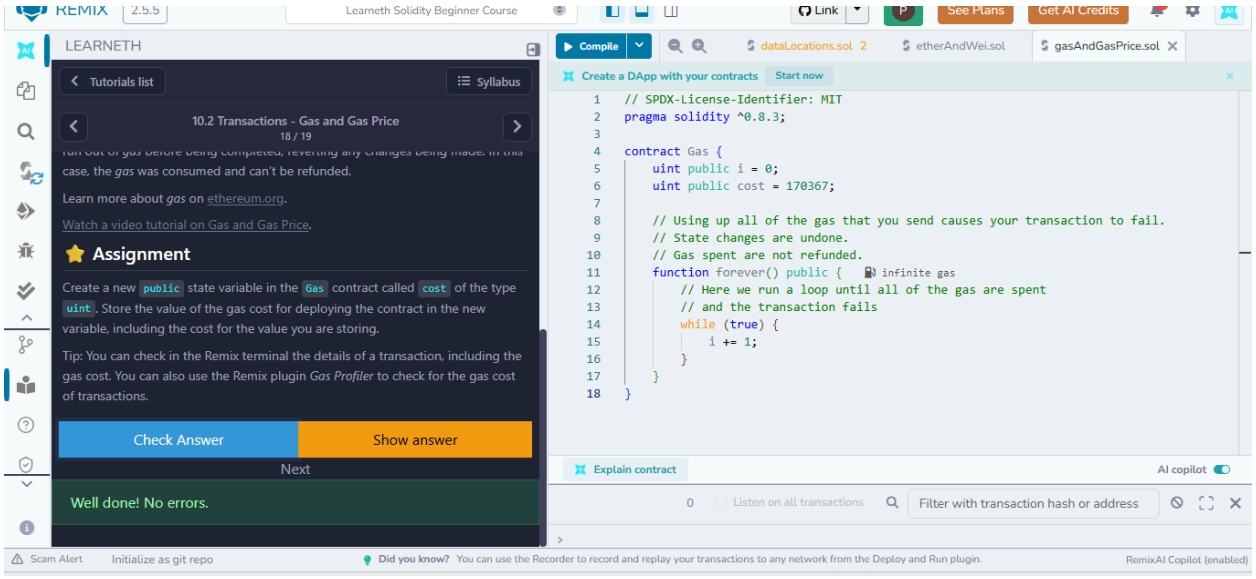
Tutorial-09 Data Locations:



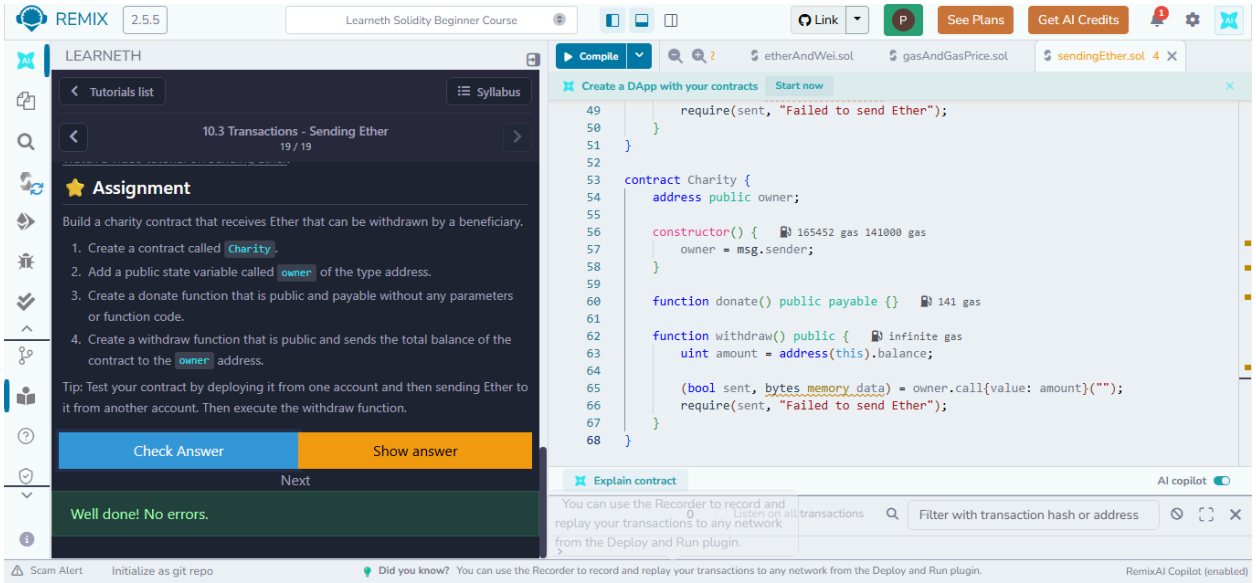
Tutorial-10.1 Transactions- Ether and Wei



Tutorial-10.2 Transactions- Gas and Gas Price:



Tutorial-10.3 Transactions- Sending Ether:



5. Conclusion

The experiment was successfully completed using Remix IDE. The basic structure and constructs of Solidity were studied, and smart contracts were successfully written, compiled, deployed, and executed using Remix VM. The experiment provided practical knowledge of Solidity programming, smart contract development, and basic blockchain transaction concepts. The 19 assignments from the Solidity Basics LearnETH course were also completed successfully.